

Fig. 7: An actual-size, single-side PCB for wireless robot

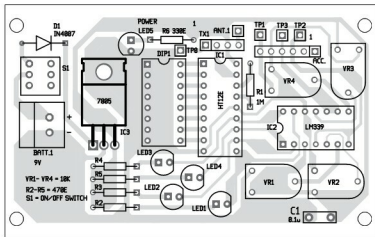


Fig. 8: Component layout for the PCB in Fig. 7

minimises assembly errors. Carefully assemble the components and double-check for any overlooked error. Use IC bases for the ICs. Before inserting the microcontroller and other ICs in their bases, check the correct supply voltage. Suitable connectors are provided on the PCB to connect geared motors M1 and M2. Connect the motor and power the circuit with a 12V battery.

Place the gesture transmitter on

your hand as shown in Fig. 1. Using presets VR1 through VR4, adjust its sensitivity as per your liking. You can now make hand gestures as shown in Fig. 1 to move the robot in all directions.

To test the circuit, check voltages as per the test-point table.

Software

The source program is written in 'C'

language and compiled using Keil μ Vision4 compiler to generate the Intel hex code. The generated hex code is burnt into the microcontroller using a suitable programmer. It is well-commented and easy to understand. ●

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